

CY 704: Introduction to Theoretical Chemistry

Course Code: CY 704

Course Name: Introduction to Theoretical Chemistry

L-T-P-C: 3-0-0-3

Elective or core: Elective

Students intended for: Ph.D.

Prerequisite: M.Sc. Semester: Odd

Approval: 2nd Senate

Course contents

- The Basics of Quantum Mechanics, Model Problems, Exact Solution. [6 Lectures]
- Understanding Energy surfaces, beyond model problems, normal modes, local modes, transition states, symmetry. [6 Lectures]
- An overview of spectroscopy, theoretical chemistry: Structure, bonding, rate of changes, Molecular [6 Lectures]
- Electronic structure: Structure atomic and molecular orbitals, Experimental Probes of Electronic Structure [6 Lectures]
- Statistical Mechanics: Collections of Molecules at or Near Equilibrium, Monte Carlo Evaluation of Properties, Molecular Dynamics Simulations, applications. [10 Lectures]
- Chemical Dynamics: Theoretical Treatment of Chemical Change and Dynamics, Experimental Probes of Reaction Dynamics. [8 Lectures]

Text & Reference Books:

1. Jack Simons, **An Introduction to Theoretical Chemistry**, Cambridge University Press, 2003.
2. C. J. Cramer, **Essentials of Computational Chemistry**, Wiley, 2002.
3. R. D. Levine and R. B. Bernstein, **Molecular Reaction Dynamics and Chemical Reactivity**, Oxford University Press, 1997.
4. A. R. Leach, **Molecular Modeling**, 2nd edition, Prentice Hall, 2001.